Universidad Modelo
Escuela de Ingeniería

Evidencias de Implementación: Azure App Service y Container Registry

Ingeniería en Desarrollo de Tecnología y Software

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1. **Descripción del Servicio**

Para la lógica de servidor, se implementó una arquitectura basada en contenedores Docker. El flujo consiste en empaquetar la aplicación en una imagen, almacenarla en un registro privado **(ACR)** y ejecutarla en un **App Service de Linux**.

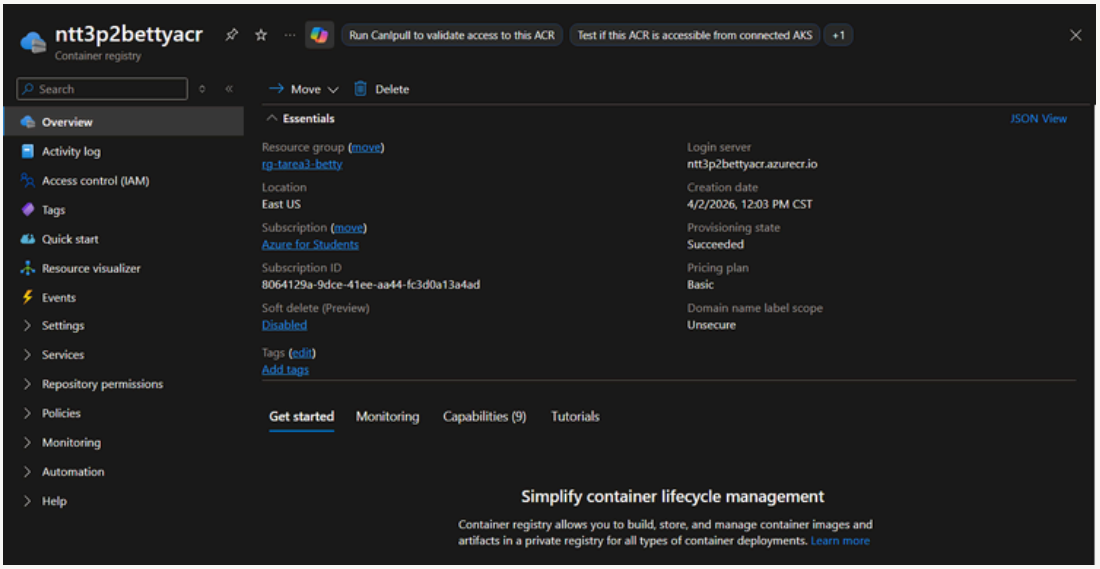
2. **Configuración de Infraestructura y Registro**

Se crearon dos recursos principales mediante Terraform:

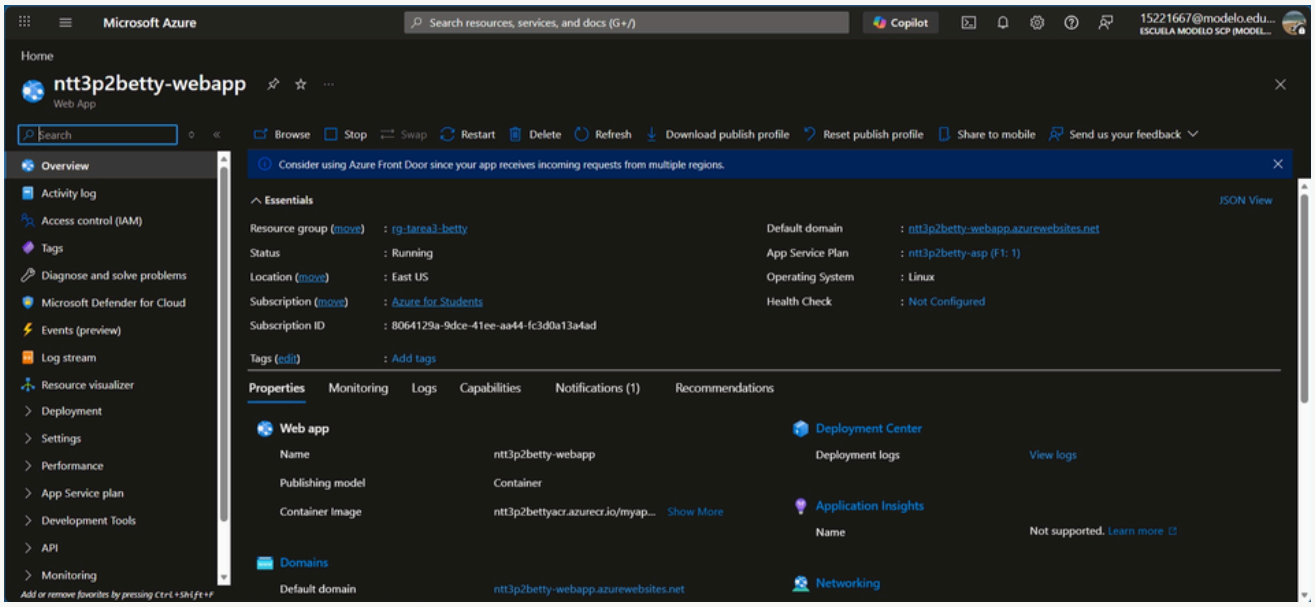
1. **Azure Container Registry (ACR):** Almacén privado para la imagen Docker.
2. **App Service Plan & Web App:** Entorno de ejecución configurado para contenedores.

NuevasTecnologias-Tarea3P2 > iac > acr.tf

```
resource "azurerm_container_registry" "acr" {
  name                = "${var.prefix}acr"
  resource_group_name = azurerm_resource_group.rg.name
  location            = azurerm_resource_group.rg.location
  sku                 = "Basic"
  admin_enabled       = true
}
```



```
NuevasTecnologias-Tarea3P2 > iac > appservice.tf
resource "azurerm_service_plan" "asp" {
  name                = "${var.prefix}-asp"
  location            = azurerm_resource_group.rg.location
  resource_group_name = azurerm_resource_group.rg.name
  os_type             = "Linux"
  sku_name            = "F1"
}

resource "azurerm_linux_web_app" "app" {
  name                = "${var.prefix}-webapp"
  location            = azurerm_resource_group.rg.location
  resource_group_name = azurerm_resource_group.rg.name
  service_plan_id     = azurerm_service_plan.asp.id

  site_config {
    always_on = false # F1 SKU doesn't support always_on
    application_stack {
      docker_image_name   = "${azurerm_container_registry.acr.login_server}/${var.docker_image_name}:${var.docker_image_tag}"
      docker_registry_url = "https://${azurerm_container_registry.acr.login_server}"
      docker_registry_username = azurerm_container_registry.acr.admin_username
      docker_registry_password = azurerm_container_registry.acr.admin_password
    }
  }

  app_settings = {
    "DOCKER_REGISTRY_SERVER_URL"      = "https://${azurerm_container_registry.acr.login_server}"
    "DOCKER_REGISTRY_SERVER_USERNAME" = azurerm_container_registry.acr.admin_username
    "DOCKER_REGISTRY_SERVER_PASSWORD" = azurerm_container_registry.acr.admin_password
  }
}
```

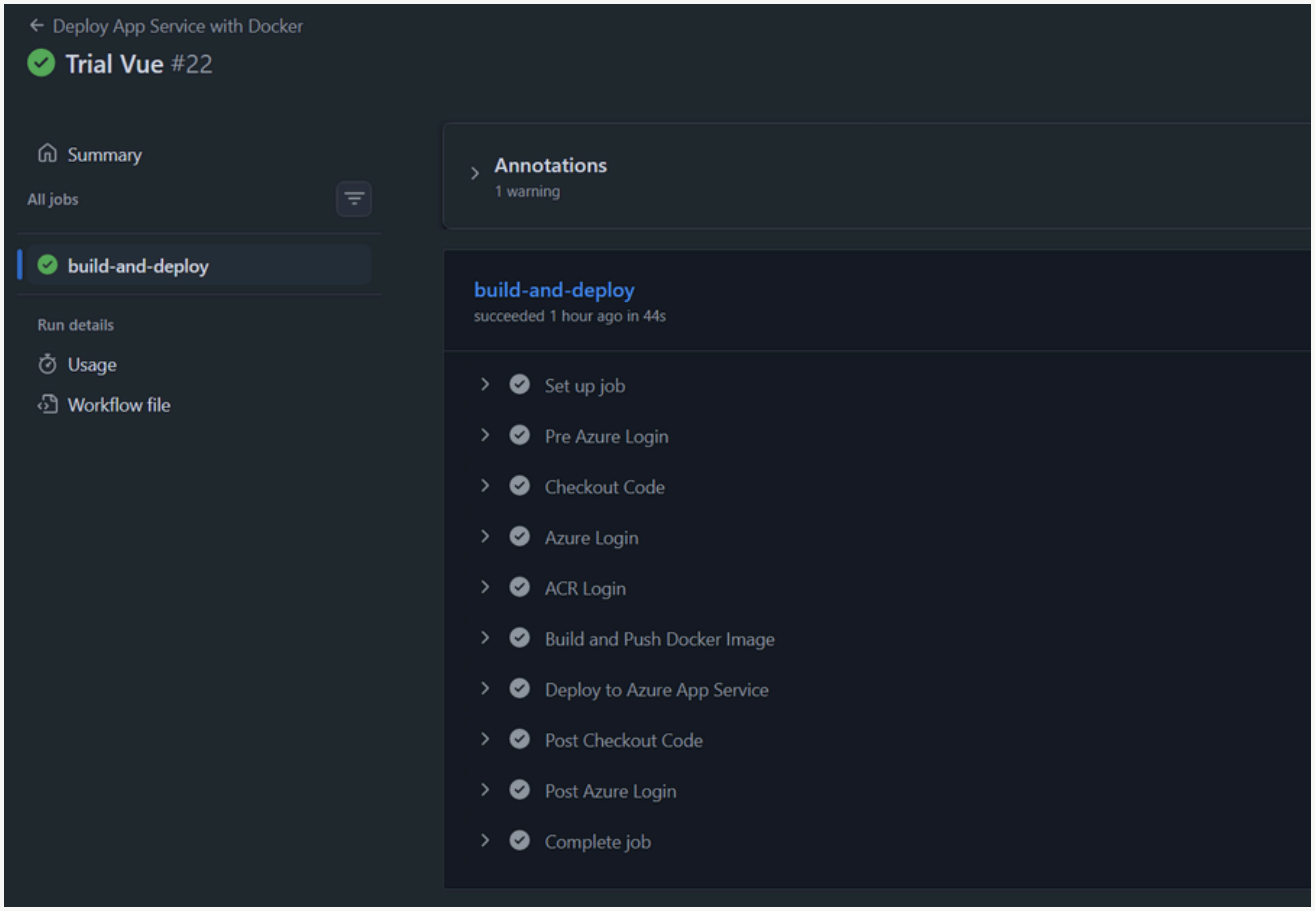


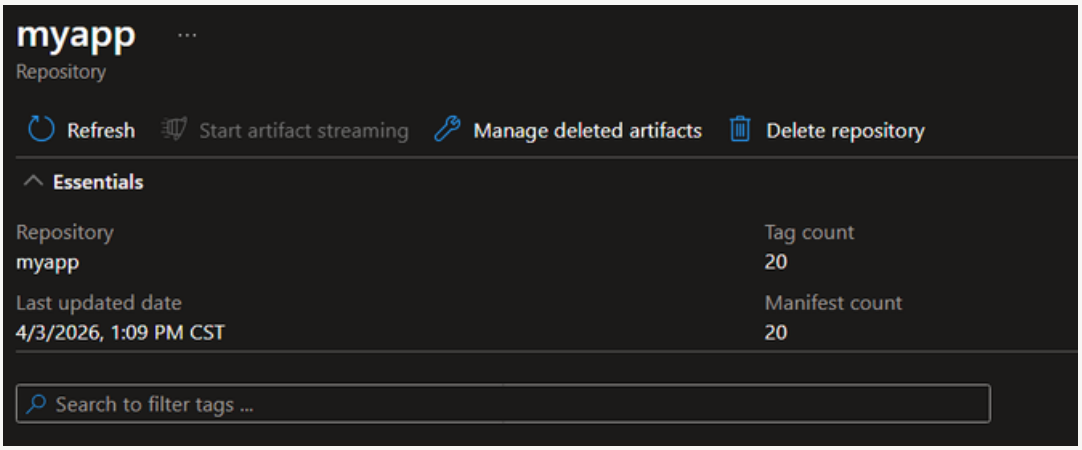
3. **Pipeline de Construcción y Push**

El flujo de trabajo automatiza el docker build utilizando el Dockerfile ubicado en /app. Posteriormente, realiza el login en Azure y hace el push de la imagen al registro.

NuevasTecnologias-Tarea3P2 > .github > workflows > appservice.yml > ...

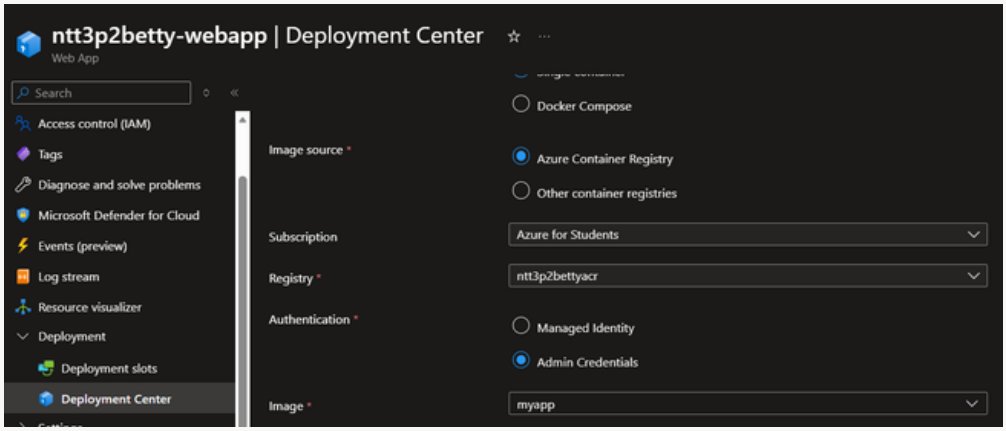
```yaml
on:
  push:
  workflow_dispatch:

jobs:
  build-and-deploy:
    runs-on: ubuntu-latest
    defaults:
      run:
        working-directory: ./app
    steps:
      - name: Checkout Code
        uses: actions/checkout@v3

      - name: Azure Login
        uses: azure/login@v1
        with:
          creds: ${{ secrets.AZURE_CREDENTIALS }}

      - name: ACR Login
        uses: azure/docker-login@v1
        with:
          login-server: ${{ vars.ACR_NAME }}.azurecr.io
          username: ${{ vars.ACR_NAME }}
          password: ${{ secrets.ACR_PASSWORD }}

      - name: Build and Push Docker Image
        run: |
          docker build -t ${{ vars.ACR_NAME }}.azurecr.io/${{ vars.DOCKER_IMAGE_NAME }}:${{ github.sha }} .
          docker push ${{ vars.ACR_NAME }}.azurecr.io/${{ vars.DOCKER_IMAGE_NAME }}:${{ github.sha }}

      - name: Deploy to Azure App Service
        uses: azure/webapps-deploy@v2
        with:
          app-name: ${{ vars.APP_SERVICE_NAME }}
          publish-profile: ${{ secrets.AZURE_WEBAPP_PUBLISH_PROFILE }}
          images: ${{ vars.ACR_NAME }}.azurecr.io/${{ vars.DOCKER_IMAGE_NAME }}:${{ github.sha }}
```

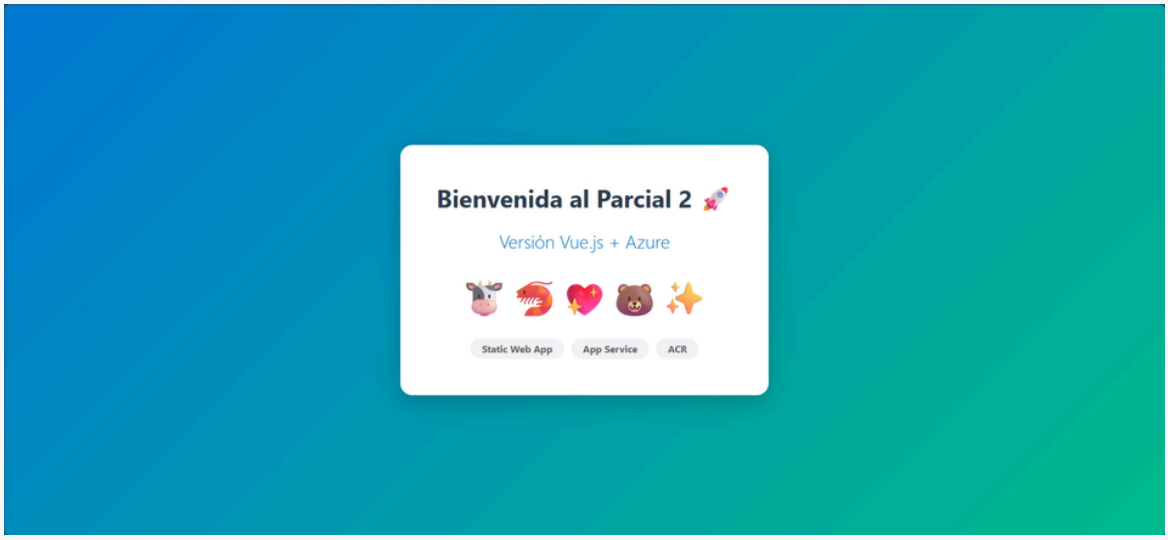




4. **Despliegue y Validación Final**

Se confirma que el App Service está ejecutando la imagen más reciente proveniente del registro privado: https://ntt3p2betty-webapp.azurewebsites.net/





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